Wetting-angle-kit: a Python package to streamline the computation of wetting contact angles of nanodroplets on surfaces

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Summary

Wetting-angle-kit is a Python toolkit designed to extract wettability properties, specifically the contact angle of a droplet on a surface, from molecular dynamics (MD) simulations. The software is intended for researchers working in molecular simulation of interfaces between liquids and solid surfaces.

It supports a variety of standard file formats including extended XYZ, LAMMPS, and ASE-readable trajectories and offers two distinct computational methods for contact angle analysis. Furthermore, the package includes robust utilities for statistical post-processing and data visualization, providing a comprehensive workflow for wettability studies. This integrated approach reduces the need for custom analysis scripts and improves reproducibility across different simulation setups.

Statement of need

The computation of contact angles from MD simulations has advanced significantly since early methodologies were proposed in 1997, with notable developments occurring in 2012, 2016, and 2024 (Hautman, Halley, and Rhee 1997; Rafiee et al. 2012; Vega and Coninck 2016; Carlson et al. 2024). Despite these advancements, the field currently lacks a standardized, unified tool for comparing and validating the diverse methods used to derive contact angles. This fragmentation poses challenges with respect to reproducibility and collaborative research. These implementations are often not publicly available or lack sufficient documentation, further limiting reproducibility. In addition, the lack of a standardized framework makes it difficult to benchmark different approaches or assess the impact of methodological choices.

Wetting-angle-kit addresses this gap by providing a flexible, open-source package. It allows researchers to implement new post-processing methods for contact angle analysis, benchmark them against established techniques, and establish a consistent baseline for wettability analysis in MD.

State of the field

General-purpose molecular simulation post-processing tools, such as OVITO (Stukowski 2010), MDSuite (Tovey et al. 2023), and MDAnalysis (Gowers et al. 2016; Michaud-Agrawal et al. 2011), provide flexible frameworks for analyzing and visualizing trajectories. However, they do not include a standardized implementation of contact angle extraction methods, which are typically developed as custom scripts tailored to specific systems.

Existing approaches to estimating the contact angle vary significantly, ranging from geometric fitting techniques to density-based methods, and often offer limited interoperability and comparability. Wetting-angle-kit complements existing tools by focusing specifically on wettability analysis and providing a consistent environment for comparing multiple methods. This design promotes reproducibility and facilitates the development and/or implementation of other methods.

Software design

Wetting-angle-kit is organized into three main components: parsers, contact angle computation methods, and visualization. This modular design separates data handling, analysis, and visualization, allowing each component to evolve independently and simplifying the integration of new features.

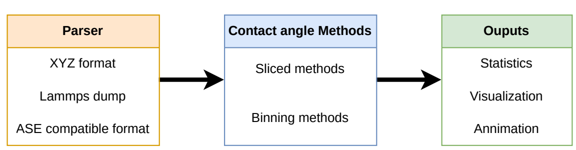

Figure 1: Package structure.

The parser module provides a unified interface for reading trajectory data from multiple formats, ensuring consistent handling of atomic coordinates, simulation boxes, and frame information. This abstraction ensures that analyses are independent of the input format, enabling consistent workflows across different simulation engines.The parser relies on established trajectory-reading tools when available, while extended XYZ parsing is implemented directly within the package.The parser also consistently handles periodic boundary conditions, ensuring that droplet shapes are correctly reconstructed across simulation boundaries and avoiding artifacts in interface detection.

This consistency facilitates seamless integration with downstream analysis methods and ensures the system's scalability, enabling researchers to easily incorporate support for additional file formats or simulation engines.

The contact angle computation methods (analysis) module implements two complementary approaches for contact angle estimation (Fig. 2).

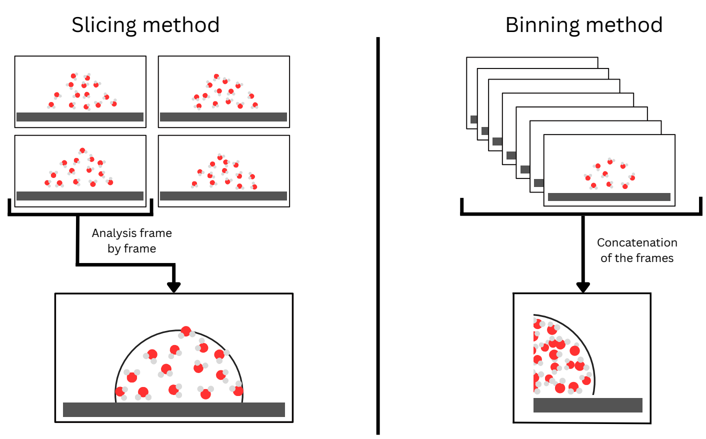

Figure 2: Schema of the two contact angle analysis methods.

The slicing method performs frame-by-frame geometric analysis, enabling detailed temporal resolution at the cost of higher computational expense. In practice, this approach provides a local characterization of the liquid–vapor interface, allowing the detection of asymmetries and transient deformations of the droplet shape. It is particularly well suited for non-equilibrium simulations or systems where the droplet deviates from an ideal spherical cap. In contrast, the binning method constructs time-averaged density fields, providing a computationally efficient approach suitable for large datasets and symmetric systems. By averaging particle positions over time, this method reduces thermal fluctuations and produces a smoother and more stable interface. It is therefore particularly effective for extracting equilibrium contact angles from noisy datasets. However, this temporal averaging may obscure short-lived fluctuations and local deviations from ideal geometries. These two approaches reflect a trade-off between temporal resolution and statistical robustness, allowing users to select the method best suited to their system.

Additionally, wetting-angle-kit supports two geometric models commonly used in the literature: spherical and cylindrical (Scocchi et al. 2011) (Fig. 3). While spherical droplets provide a more direct representation of droplet curvature, cylindrical geometries reduce curvature effects and computational cost, at the expense of relying on an idealized geometry.

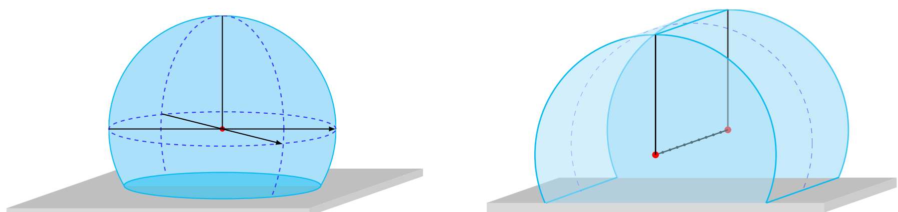

Figure 3: Geometric representations of droplets used in the analysis: spherical droplet (left) and cylindrical droplet (right).

The software architecture relies on abstract base classes to enforce consistent interfaces and facilitate extensibility. This design enables users to implement new computational methods while maintaining compatibility with existing workflows, promoting reuse and method comparison. It also facilitates the integration of newly developed methods into an existing and standardized analysis pipeline.

Visualization tools are included to support interpretation and validation of analysis results without requiring external post-processing tools. These tools provide representations of droplet geometries, enabling users to directly inspect the quality of interface detection and fitting procedures. By facilitating visual verification of intermediate and final results, they help identify potential artifacts or inconsistencies in the analysis and improve the reliability of extracted contact angles.

Research impact statement

Wetting-angle-kit provides a reproducible framework for contact angle analysis in molecular simulations, addressing a common need in studies of nanoscale wetting. The package has been validated using MD simulations of water droplets on graphene and polymer substrates, yielding contact angle values consistent with literature results (e.g., ~93° for graphene, ~110° for PTFE), see Fig. 4. The reported contact angles were obtained by analyzing droplets of increasing size and extrapolating to the macroscopic limit using the Modified Young's relation, where the contact angle is related to droplet size through a line-tension correction term, enabling estimation of the infinite-droplet contact angle. These results are consistent with literature values obtained using similar carbon-oxygen LJ parameters (Jorgensen, Maxwell, and Tirado-Rives 1996).

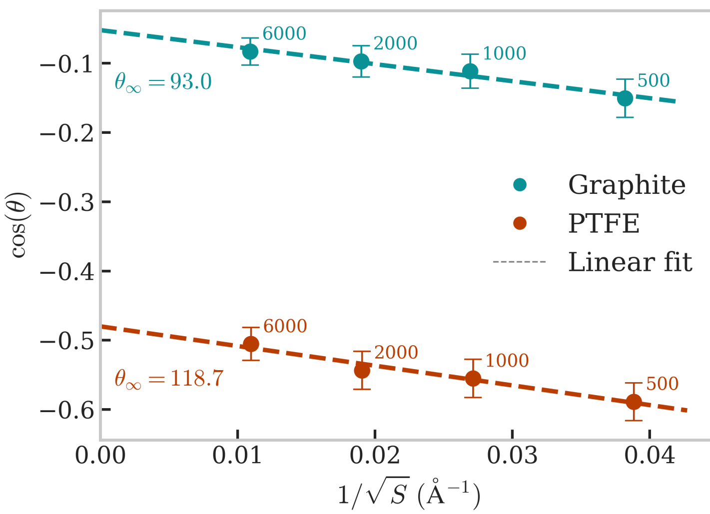

Figure 4: Figure 4: Size-dependent contact angle analysis for water droplets on graphite and PTFE substrates. Values of $\cos(\theta)$ are plotted as a function of the inverse square root of the droplet surface area for droplets containing between 500 and 6000 water molecules. Linear extrapolation following the Modified Young's relation is used to estimate the macroscopic (infinite-droplet) contact angle.

By enabling systematic comparison of analysis methods and providing standardized workflows, the software supports more robust and reproducible wettability studies. Its modular design also facilitates integration into existing simulation pipelines and encourages community-driven extensions. The package is expected to be particularly useful for researchers using various types of force fields (classical and MLIPs) or investigating nanoscale interfacial phenomena.

AI usage disclosure

Generative AI tools were used in the development of the software, for drafting and assisting debugging. Generative AI was used to assist in refining the language, traduction and clarity of the manuscript and docstring. All AI-assisted contributions were verified and approved by the authors.

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